

Saddle-Field Policy Optimization for Zero-Sum Markov Games: Adaptive Lyapunov-Drift Control with Finite-Time Stationarity Guarantees

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Abstract—Two-player zero-sum Markov games model robust reinforcement learning and competitive policy optimization. Rotational components in their saddle fields can hinder first-order updates. We identify the saddle field $F(z)$ over the joint policy parameter z and a Jacobian-induced curvature direction $G(z) = \nabla F(z)F(z)$, where ∇F is the field Jacobian, for rotational coupling. A symmetric–antisymmetric Jacobian decomposition characterizes when the curvature direction decreases field energy. We then develop a geometry-aware controller that assigns separate step sizes to the two directions. Minimizing a local model of the one-step change in a composite Lyapunov function yields the two step sizes through a box-constrained quadratic program (QP). Under local geometric coverage, the controller attains an $O(1/K)$ bound over K joint updates on the time-average expected stationarity residual, plus an explicit residual floor. A positive-semidefinite symmetric part and nonsingularity of the metric-transformed Jacobian at the target stationary point are shown to imply neighborhood coverage, including the nonsingular pure-rotation case. Under a local policy-space error bound, this stationarity result transfers to guarantees on the unregularized Nash gap and the maximizing player’s worst-case-return loss. Experiments across four settings validate the predicted geometry; at the final checkpoint of every Markov-game task, the controller has higher mean worst-case return and lower mean exploitability than the matched field-only ablation, including a 79.1% exploitability reduction on neural *CyclicControl*.

Index Terms—Zero-sum Markov games, robust reinforcement learning, stochastic saddle fields, Lyapunov drift, adaptive step size.

I. INTRODUCTION

A TWO-PLAYER zero-sum Markov game describes a sequential interaction in which two decision makers optimize opposing objectives [1], [2]. The state records the information needed for the next decision, such as a controlled system’s physical configuration, operating condition, or interaction history. Each player follows a policy that maps the current state to a distribution over its available actions. Their joint action determines both the instantaneous payoff and the next-state distribution, and the players respectively maximize and minimize the same expected cumulative return. Key applications include robust control under uncertain dynamics, adversarial reinforcement learning (RL), and competitive multi-agent decision making, where the maximizing policy selects

control or decision actions and the minimizing policy represents uncertainty, disturbances, or an adaptive opponent [3]–[8].

Policy optimization in a zero-sum Markov game is more involved than in a single-agent Markov decision process (MDP). An MDP optimizes one policy under fixed reward and transition mechanisms, whereas both policies in a Markov game adapt and jointly determine the state distribution, return, and policy gradients. An update by either player therefore changes the objective encountered by the other and produces coupled ascent–descent dynamics rather than a single policy-improvement direction. Neural policy parameterization further yields a generally nonconvex–nonconcave objective, while trajectory sampling introduces noise and bias into its gradient observations. The Jacobian of the joint policy dynamics can also contain a large antisymmetric component that rotates the update around a stationary point. Consequently, simultaneous gradient descent–ascent (GDA) can cycle or diverge even with small step sizes; related stability effects arise in linear–quadratic robust RL [9].

Classical dynamic-programming and value-based methods solve finite games through Bellman–Shapley operators or min–imax value updates [1], [2], [5], [6]. Their computational and storage requirements motivate parameterized, trajectory-based policy optimization in larger state and action spaces. Policy-gradient, actor–critic, optimistic, and natural-gradient schemes update the two policies directly and provide convergence or sample-complexity guarantees under different information and regularity models [10]–[14]. These methods establish policy optimization as a scalable approach to Markov games, while the interaction between simultaneous player updates remains central to their stability. Competitive policy optimization incorporates this interaction through a local bilinear model of the two objectives [15].

Proximal and extrapolation methods provide a complementary route for controlling coupled saddle dynamics. The proximal point method (PPM), the extragradient method (EGM), Mirror-Prox, and policy extragradient introduce corrections that can suppress rotational motion under suitable regularity and step-size conditions [16]–[20]. Local expansions identify a Jacobian-induced term as the leading correction shared by PPM and EGM beyond the GDA field direction [19], [21]. Variable-step-size extragradient methods tune the extrapolation and update step sizes from observed operator information [22], [23], while differentiable-game methods construct Hamiltonian, consensus, or symplectic adjustments from the game Jacobian [24]–[27]. These developments expose

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curvature as a mechanism for controlling rotation, and they leave a concrete policy-optimization question: how should the field and curvature contributions be balanced as the local game geometry and stochastic oracle quality change along training?

We develop a geometry-adaptive stochastic policy optimizer that answers this question through two search directions with independently selected step sizes. The saddle-field analysis supplies a first-order field direction and a Jacobian-induced curvature direction, and an exact symmetric–antisymmetric decomposition characterizes when the curvature direction decreases field energy. At each joint policy update, stochastic observations of these directions are combined with two independently selected nonnegative step sizes. A composite Lyapunov function measures both parameter-space stationarity and the policy-space improvement available through unilateral deviations. The controller selects the two step sizes by minimizing a second-order local model of the conditional Lyapunov drift over a two-dimensional box. Thus, the saddle geometry determines the search directions, whereas the Lyapunov-drift model determines their adaptive step sizes. The construction extends Lyapunov-drift step-size design for stochastic optimization [28], [29] to a two-direction saddle-field update and yields a closed-form box-constrained quadratic program (QP).

The main contributions of this paper are summarized as follows.

- **Adaptive two-direction policy optimization.** We independently select the step sizes assigned to the field and curvature directions through a two-variable box QP with a closed-form solution that minimizes a local model of composite Lyapunov drift. The composite Lyapunov function combines first-order stationarity with regularized policy-space performance on a common scale; the QP dominates every feasible field-only, curvature-only, or prescribed-coupling update in the same box, with a reduced-gradient condition for strict improvement.
- **Geometric characterization of the curvature direction.** We show that the curvature correction shared by PPM and EGM decreases field energy exactly when the antisymmetric Jacobian action exceeds the symmetric action. The resulting criterion identifies the local rotational geometry that motivates adaptive curvature use.
- **Finite-time guarantees.** For biased stochastic policy oracles, two-direction geometric coverage yields an $O(1/K)$ transient in the time-average expected stationarity residual over K joint policy updates, together with an explicit stochastic error floor. We establish neighborhood coverage from a positive-semidefinite symmetric Jacobian part and nonsingularity of the metric-transformed Jacobian at the target, including the nonsingular pure-rotation case. Under a local policy-space error bound, the stationarity guarantee further yields bounds on the ergodic Nash gap and the maximizing player’s worst-case-return loss.
- **Evidence across four levels of policy optimization.** Controlled normal-form experiments verify the predicted curvature geometry, while tabular, neural-policy, and finite-trajectory Markov-game experiments evaluate optimization and policy performance. Across the reported Markov-game tasks, the adaptive two-direction update raises final mean

worst-case return and reduces final mean exploitability relative to its matched field-only ablation.

The remainder of the paper develops the argument as follows. Section II formulates the parameterized Markov game, derives the two search directions, and introduces their stochastic recursion. Section III constructs the Lyapunov-drift step-size controller, Section IV establishes the finite-time and policy-space guarantees, Section V gives the practical implementation, and Section VI presents the experiments.

II. ZERO-SUM MARKOV GAMES AND STOCHASTIC TWO-DIRECTION POLICY DYNAMICS

A. Zero-Sum Markov Game and Saddle-Field Formulation

We consider an infinite-horizon discounted two-player zero-sum Markov game with state space $\mathcal{S}$, finite maximizing- and minimizing-player action spaces $\mathcal{A}$ and $\mathcal{B}$, transition kernel P , bounded reward r to the maximizing player, discount factor $\alpha \in (0, 1)$, and initial-state distribution ρ . For a maximizing-player policy π and a minimizing-player policy ν , the maximizing player’s discounted return is

$$J(\pi, \nu) \triangleq \mathbb{E}_{\rho, \pi, \nu} [\sum_{t=0}^{\infty} \alpha^t r(s_t, a_t, b_t)].$$

Here s_t , a_t , and b_t denote the state, maximizing-player action, and minimizing-player action at time t , respectively. The expectation $\mathbb{E}_{\rho, \pi, \nu}$ is taken over the trajectory induced by the initial-state distribution, the two policies, and the transition kernel. The policy-space saddle objective is $\sup_{\pi} \inf_{\nu} J(\pi, \nu)$. Define the discounted causal entropies

$$\begin{aligned} H_{\alpha}^{\pi}(\pi, \nu) &\triangleq \mathbb{E}_{\rho, \pi, \nu} \left[\sum_{t \geq 0} \alpha^t (-\log \pi(a_t | s_t)) \right], \\ H_{\alpha}^{\nu}(\pi, \nu) &\triangleq \mathbb{E}_{\rho, \pi, \nu} \left[\sum_{t \geq 0} \alpha^t (-\log \nu(b_t | s_t)) \right], \end{aligned}$$

with $0 \log 0 = 0$. For entropy weight $\varepsilon \geq 0$, the corresponding regularized return is

$$J_{\varepsilon}(\pi, \nu) \triangleq J(\pi, \nu) + \varepsilon H_{\alpha}^{\pi}(\pi, \nu) - \varepsilon H_{\alpha}^{\nu}(\pi, \nu). \quad (1)$$

The opposite signs preserve the max–min convention while encouraging stochastic policies for both players. Setting $\varepsilon = 0$ recovers the unregularized return J . We use J_{ε} to construct the policy-update field and retain J for the final unregularized performance measures. In large finite or continuous state spaces, we adopt the standard parameterized formulation because exact policy-space optimization is generally intractable. The maximizing-player policy is represented by $\pi_{\theta}(a | s)$ with $\theta \in \mathbb{R}^{d_{\theta}}$, and the minimizing-player policy by $\nu_{\psi}(b | s)$ with $\psi \in \mathbb{R}^{d_{\psi}}$, giving

$$\begin{aligned} J_{\varepsilon}(\theta, \psi) &\triangleq J_{\varepsilon}(\pi_{\theta}, \nu_{\psi}), \\ \max_{\theta} \min_{\psi} J_{\varepsilon}(\theta, \psi). \end{aligned}$$

Write $\mathbf{z} \triangleq (\theta, \psi) \in \mathbb{R}^d$, where $d \triangleq d_{\theta} + d_{\psi}$. The joint policy ascent–descent dynamics define the saddle field

$$\mathbf{F}(\mathbf{z}) \triangleq \begin{bmatrix} -\nabla_{\theta} J_{\varepsilon}(\theta, \psi) \\ \nabla_{\psi} J_{\varepsilon}(\theta, \psi) \end{bmatrix}. \quad (2)$$

The first-order stationary set is $\mathcal{Z}^{\star} \triangleq \{\mathbf{z} : \mathbf{F}(\mathbf{z}) = \mathbf{0}\}$, where $\mathbf{0}$ is the zero vector. Every $\mathbf{z}^{\star} \in \mathcal{Z}^{\star}$ satisfies the unconstrained

first-order Karush–Kuhn–Tucker (KKT) conditions for the parameterized objective J_ε . Under neural policy parameterization, $\mathcal{Z}^*$ is the parameter-space first-order stationary set. This distinction matters because a nonconvex or noninjective policy parameterization can satisfy $\mathbf{F}(\mathbf{z}) = \mathbf{0}$ even when a player can still improve by taking a best response in the full policy space. The policy-space results in Section IV-C use the Nash gap to measure this remaining unilateral-improvement incentive and additionally bound the maximizing player’s worst-case-return loss.

B. Classical Saddle Updates and the Curvature Direction

The saddle-field formulation exposes the rotational component that can destabilize first-order updates. To understand how established saddle methods counter this effect, we examine the Jacobian-based correction generated by PPM and EGM. For a scalar step size $s > 0$, let $\mathbf{z}_+$ denote the next joint policy parameter. The representative GDA, PPM, and EGM updates are

$$\begin{aligned} \text{GDA: } \mathbf{z}_+^{\text{GDA}} &= \mathbf{z} - s\mathbf{F}(\mathbf{z}), \\ \text{PPM: } \mathbf{z}_+^{\text{PPM}} &= \mathbf{z} - s\mathbf{F}(\mathbf{z}_+^{\text{PPM}}), \\ \text{EGM: } \tilde{\mathbf{z}} &= \mathbf{z} - s\mathbf{F}(\mathbf{z}), \quad \mathbf{z}_+^{\text{EGM}} = \mathbf{z} - s\mathbf{F}(\tilde{\mathbf{z}}), \end{aligned} \quad (3)$$

where $\tilde{\mathbf{z}}$ is the EGM extrapolation point. GDA is explicit, PPM is implicit, and EGM uses an extrapolated field query.

The following assumption states the local regularity required by the template expansion and one-step analysis.

Assumption 1 (Local smoothness). *There exists an open set $\mathcal{D} \subset \mathbb{R}^d$ such that: (a) all iterates and probing points used in the analysis lie in $\mathcal{D}$; (b) $J_\varepsilon : \mathbb{R}^d \rightarrow \mathbb{R}$ is four times continuously differentiable (C^4) on $\mathcal{D}$; (c) all derivatives of J_ε up to order four are bounded on $\mathcal{D}$.*

Conditions (b)–(c) imply that $\mathbf{F}$ is three times continuously differentiable, i.e. $\mathbf{F} \in C^3(\mathcal{D}; \mathbb{R}^d)$; after prescribing an implementable region $\mathcal{T} \subset \mathcal{D}$, the safeguards of Section V can maintain Condition (a). Taylor expansion of the field uses its Jacobian $\nabla \mathbf{F}$ and second-derivative tensor $\nabla^2 \mathbf{F}$, both taken with respect to $\mathbf{z}$. The remainder bounds use the Euclidean norm $\|\cdot\|$ and its induced operator and tensor norms.

Proposition 1 (Discrete-time second-order template expansions). *Suppose Assumption 1 holds and fix $\mathbf{z} \in \mathcal{D}$. There exists $\bar{s} > 0$ such that, for every $s \in (0, \bar{s}]$, the PPM equation has a unique local solution, the EGM extrapolation and update points lie in $\mathcal{D}$, and*

$$\begin{aligned} \text{GDA: } \mathbf{z}_+ - \mathbf{z} &= -s\mathbf{F}(\mathbf{z}), \\ \text{PPM: } \mathbf{z}_+ - \mathbf{z} &= -s\mathbf{F}(\mathbf{z}) + s^2\nabla\mathbf{F}(\mathbf{z})\mathbf{F}(\mathbf{z}) + R_s^{\text{PPM}}(\mathbf{z}), \\ \text{EGM: } \mathbf{z}_+ - \mathbf{z} &= -s\mathbf{F}(\mathbf{z}) + s^2\nabla\mathbf{F}(\mathbf{z})\mathbf{F}(\mathbf{z}) + R_s^{\text{EGM}}(\mathbf{z}), \end{aligned}$$

The remainders satisfy $\|R_s^{\text{PPM}}(\mathbf{z})\| \leq C_{\text{PPM}}s^3$ and $\|R_s^{\text{EGM}}(\mathbf{z})\| \leq C_{\text{EGM}}s^3$ for finite constants depending only on local bounds of $\mathbf{F}$, $\nabla\mathbf{F}$, and $\nabla^2\mathbf{F}$ on $\mathcal{D}$.

Unlike GDA, PPM and EGM share the leading-order correction $+s^2\nabla\mathbf{F}(\mathbf{z})\mathbf{F}(\mathbf{z})$, which can suppress rotational saddle dynamics at suitable step sizes. This contrast motivates writing $\mathbf{A}(\mathbf{z}) \triangleq \nabla\mathbf{F}(\mathbf{z})$ and isolating the curvature direction

$$\mathbf{G}(\mathbf{z}) \triangleq \mathbf{A}(\mathbf{z})\mathbf{F}(\mathbf{z}). \quad (4)$$

The expansion identifies the curvature direction shared by PPM and EGM, and the next step is to characterize how this direction changes the local saddle dynamics.

C. Curvature Descent and Independent Step Sizes

Proposition 1 identifies $+\mathbf{G}$ as the leading correction shared by PPM and EGM. Its effect on the stationarity residual is governed by the local Jacobian geometry, which motivates characterizing when $+\mathbf{G}$ decreases the Euclidean field energy

$$\phi(\mathbf{z}) \triangleq \frac{1}{2} \|\mathbf{F}(\mathbf{z})\|^2,$$

which is nonnegative and vanishes exactly on $\mathcal{Z}^*$.

Fig. 1 first illustrates the two directions on the bilinear max–min saddle $L(x, y) = -xy$. Its field is $\mathbf{F} = J_0\mathbf{z}$, where

$$J_0 \triangleq \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

Since $J_0^2 = -\mathbf{I}_2$, where $\mathbf{I}_2$ is the 2×2 identity matrix, the field direction $-\mathbf{F}$ is tangent to the level sets of ϕ , whereas $+\mathbf{G} = J_0^2\mathbf{z} = -\mathbf{z} = -\nabla\phi$ points inward. This geometry explains why GDA rotates while PPM and EGM contract at the displayed step size in panel (a), and panel (b) shows the two directional roles directly. Panel (c) further shows that changing the curvature-to-field step-size ratio changes both contraction and residual rotation, motivating separate selection of the two step sizes.

To extend this observation beyond the bilinear game, fix $\mathbf{z} \in \mathcal{D}$, write $\mathbf{A} = \nabla\mathbf{F}(\mathbf{z})$, and decompose the Jacobian as

$$\begin{aligned} \mathbf{A} &= \mathbf{S} + \mathbf{W}, \\ \mathbf{S} &= \text{sym}(\mathbf{A}) \triangleq \frac{1}{2}(\mathbf{A} + \mathbf{A}^\top), \\ \mathbf{W} &= \text{skew}(\mathbf{A}) \triangleq \frac{1}{2}(\mathbf{A} - \mathbf{A}^\top). \end{aligned}$$

The symmetric part $\mathbf{S}$ captures nonrotational dynamics, whereas the antisymmetric part $\mathbf{W}$ captures rotational dynamics [24]. Using $\langle \cdot, \cdot \rangle$ for the standard Euclidean inner product, the exact directional derivatives give the following criterion.

Theorem 1 (Skew-dominance characterization of curvature descent). *For every $\mathbf{z} \in \mathcal{D}$ with $\mathbf{F}(\mathbf{z}) \neq \mathbf{0}$, the field and curvature directions satisfy*

$$\langle \nabla\phi(\mathbf{z}), -\mathbf{F}(\mathbf{z}) \rangle = -\mathbf{F}(\mathbf{z})^\top \mathbf{S} \mathbf{F}(\mathbf{z}), \quad (5)$$

$$\langle \nabla\phi(\mathbf{z}), \mathbf{G}(\mathbf{z}) \rangle = \|\mathbf{S} \mathbf{F}(\mathbf{z})\|^2 - \|\mathbf{W} \mathbf{F}(\mathbf{z})\|^2. \quad (6)$$

Consequently, $+\mathbf{G}(\mathbf{z})$ strictly decreases ϕ if and only if

$$\|\mathbf{W} \mathbf{F}(\mathbf{z})\| > \|\mathbf{S} \mathbf{F}(\mathbf{z})\|. \quad (7)$$

Eq. (5) shows that the symmetric Jacobian action governs the first-order effect of the field direction. For a purely rotational field, $\mathbf{S} = \mathbf{0}$ makes $-\mathbf{F}$ tangential, while $\mathbf{W} \mathbf{F} \neq \mathbf{0}$ makes $+\mathbf{G}$ a strict descent direction. Conversely, when $\mathbf{W} = \mathbf{0}$, Eq. (6) shows that $+\mathbf{G}$ preserves or increases the field energy to first order. For a two-dimensional normal block $\mathbf{A} = \mu\mathbf{I}_2 + \sigma J_0$, the criterion reduces to $|\sigma| > |\mu|$, directly comparing rotational and radial action.

Because this balance changes with the local Jacobian geometry, we assign independent nonnegative step sizes β and γ to the two population directions:

$$\mathbf{z}_+ = \mathbf{z} - \beta\mathbf{F}(\mathbf{z}) + \gamma\mathbf{G}(\mathbf{z}), \quad \beta, \gamma \geq 0. \quad (8)$$

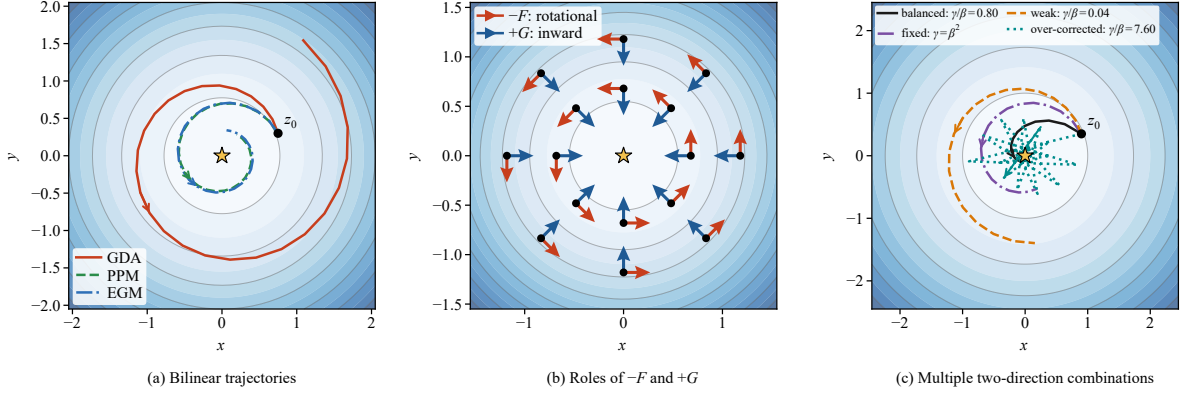


Fig. 1. Geometry of two-direction saddle updates. (a) GDA spirals outward, whereas PPM and EGM contract at the displayed step size. (b) On the bilinear game, the first-order field direction is tangential to the residual contours, whereas the shared second-order correction points inward. (c) The curvature-to-field step-size ratio controls contraction and residual rotation.

The pair $(\beta, \gamma) = (s, 0)$ recovers GDA, whereas $(\beta, \gamma) = (s, s^2)$ matches the PPM and EGM templates through second order. Eq. (8) therefore retains the classical search directions while allowing their step sizes to adapt independently to the local game geometry.

D. Stochastic Two-Direction Recursion and Oracle Model

Eq. (8) defines the joint-update family underlying the proposed recursion: the field direction retains the first-order GDA action, while the curvature direction generalizes the leading correction shared by PPM and EGM. In trajectory-based policy optimization, both population directions are observed through stochastic policy oracles. At joint update k , let $\hat{\mathbf{F}}_k$ and $\hat{\mathbf{G}}_k$ denote these observations and let $\beta_k, \gamma_k \geq 0$ denote their step sizes. The resulting implementable two-direction recursion is

$$\mathbf{z}_{k+1} = \mathbf{z}_k - \beta_k \hat{\mathbf{F}}_k + \gamma_k \hat{\mathbf{G}}_k, \quad (9)$$

Finite-trajectory policy optimization supplies such stochastic observations through sampled policy-gradient and curvature estimates [10]. For the stochastic analysis, let $\{\mathcal{F}_k\}_{k \geq 0}$ be the filtration generated by the algorithmic history revealed before the queries at joint update k .

Assumption 2 (Biased stochastic oracles). *For every k , let $\mathbf{b}_k^F, \mathbf{b}_k^G$ be $\mathcal{F}_k$ -measurable oracle biases and let $\boldsymbol{\xi}_k^F, \boldsymbol{\xi}_k^G$ be conditionally zero-mean noise variables, i.e. $\mathbb{E}[\boldsymbol{\xi}_k^F | \mathcal{F}_k] = \mathbb{E}[\boldsymbol{\xi}_k^G | \mathcal{F}_k] = \mathbf{0}$. The observed directions decompose as*

$$\hat{\mathbf{F}}_k = \mathbf{F}(\mathbf{z}_k) + \mathbf{b}_k^F + \boldsymbol{\xi}_k^F, \quad \hat{\mathbf{G}}_k = \mathbf{G}(\mathbf{z}_k) + \mathbf{b}_k^G + \boldsymbol{\xi}_k^G,$$

There exist finite $\mathcal{F}_k$ -measurable scalars $\bar{b}_k^F, \bar{b}_k^G, \sigma_{2,k}, \sigma_{3,k} \geq 0$ such that, almost surely,

- (a) $\|\mathbf{b}_k^F\| \leq \bar{b}_k^F$ and $\|\mathbf{b}_k^G\| \leq \bar{b}_k^G$;
- (b) $\mathbb{E}[\|\boldsymbol{\xi}_k^F\|^2 + \|\boldsymbol{\xi}_k^G\|^2 | \mathcal{F}_k] \leq \sigma_{2,k}^2$;
- (c) $\mathbb{E}[\|\hat{\mathbf{F}}_k\|^3 + \|\hat{\mathbf{G}}_k\|^3 | \mathcal{F}_k] \leq \sigma_{3,k}^3$.

Here $\bar{b}_k^F, \bar{b}_k^G$ are bias bounds, while $\sigma_{2,k}, \sigma_{3,k}$ bound the conditional second noise moment and third oracle moment. This model covers unbiased mini-batch estimators and bounded bias from truncated trajectories, clipped surrogates, or approximate evaluations of the curvature direction. Accordingly, $\hat{\mathbf{F}}_k$ and

$\hat{\mathbf{G}}_k$ estimate the population directions generated by J_ε , and $\mathbf{b}_k^F$ and $\mathbf{b}_k^G$ quantify their conditional estimation biases. Eq. (9) specifies the search directions and their stochastic observations. The remaining design task is to determine their two step sizes. Independent selection of β_k and γ_k allows the two step sizes to respond separately to local rotation, policy-space performance, and oracle quality. The next section therefore constructs a composite Lyapunov function and selects (β_k, γ_k) from its local one-step model.

III. GEOMETRY-AWARE LYAPUNOV-DRIFT STEP-SIZE SELECTION

Section II determines the field and curvature directions of the stochastic policy update. The remaining design problem is to select their two step sizes from the current game geometry and oracle observations. Field energy measures parameter-space stationarity, while policy-space performance requires accounting for the improvement available through unilateral deviations. We place these quantities on a common Lyapunov scale, model its conditional one-step change, and select the two step sizes from an implementable local approximation.

A. Composite Lyapunov Function

The field energy measures parameter-space stationarity. The regularized Nash gap measures unilateral policy improvement and can distinguish parameter-stationary policies that retain positive exploitability. Because these quantities have different numerical scales, the following Lyapunov function combines them: field energy guides local optimization, while the regularized gap retains policy-space performance in the step-size decision.

Definition 1 (Composite Lyapunov function). *Let $\text{diag}(\cdot)$ denote block-diagonal concatenation and let $\eta_\psi > 0$ be the relative weight of the minimizing-player gradient block. Define the block-diagonal weight matrix*

$$\mathbf{W}_V \triangleq \text{diag}(\mathbf{I}_{d_\theta}, \eta_\psi \mathbf{I}_{d_\psi}) \in \mathbb{R}^{d \times d}.$$

The subscript V identifies the metric weight used in $\mathcal{E}_F$ and $\mathcal{V}$ and distinguishes it from the antisymmetric Jacobian part $\mathbf{W} = \text{skew}(\nabla \mathbf{F})$ introduced in Section II-C. The weighted field-energy component is $\mathcal{E}_F(\mathbf{z}) \triangleq \frac{1}{2} \mathbf{F}(\mathbf{z})^\top \mathbf{W}_V \mathbf{F}(\mathbf{z})$. Its

population decrease signals along the field and curvature directions are

$$\begin{aligned}\chi_F(\mathbf{z}) &\triangleq -\langle \nabla \mathcal{E}_F(\mathbf{z}), -\mathbf{F}(\mathbf{z}) \rangle \\ &= \frac{1}{2} \mathbf{F}^\top (\mathbf{W}_V \mathbf{A} + \mathbf{A}^\top \mathbf{W}_V) \mathbf{F}, \\ \chi_G(\mathbf{z}) &\triangleq -\langle \nabla \mathcal{E}_F(\mathbf{z}), \mathbf{G}(\mathbf{z}) \rangle \\ &= -\mathbf{F}^\top \mathbf{W}_V \mathbf{A}^2 \mathbf{F}.\end{aligned}\quad (10)$$

Positive values indicate first-order decrease of the weighted field energy along the corresponding direction. For the policy-performance component, fix $\varepsilon > 0$ and write $J_\varepsilon = J_\varepsilon(\pi_\theta, \nu_\psi)$. The same regularized return that generates the saddle field defines

$$\begin{aligned}\mathcal{P}(\mathbf{z}) &\triangleq \text{Gap}_\varepsilon(\pi_\theta, \nu_\psi) \\ &= \left[\sup_{\bar{\pi}} J_\varepsilon(\bar{\pi}, \nu_\psi) - J_\varepsilon \right] + \left[J_\varepsilon - \inf_{\bar{\nu}} J_\varepsilon(\pi_\theta, \bar{\nu}) \right],\end{aligned}$$

The extrema range over stationary randomized policies, and the two brackets measure the maximizing player's regularized return improvement and the minimizing player's regularized return reduction. Thus, $\mathcal{P} \geq 0$ and vanishes when neither player can improve unilaterally; Proposition 8 gives finite-game smoothness conditions, and Proposition 7 later transfers this regularized measure to performance under J . Setting $\mathcal{P} \equiv 0$ and $\lambda_P = 0$ gives the field-energy-only controller used in the normal-form study. For fixed normalizers $\kappa_F, \kappa_P > 0$ and weights $\lambda_F > 0$, $\lambda_P \geq 0$, define the composite Lyapunov function

$$\mathcal{V}(\mathbf{z}) \triangleq \lambda_F \frac{\mathcal{E}_F(\mathbf{z})}{\kappa_F} + \lambda_P \frac{\mathcal{P}(\mathbf{z})}{\kappa_P}. \quad (11)$$

The normalizers place components with different numerical scales on a comparable basis, while the λ weights specify their relative influence on the controller.

For the recursion in Eq. (9), define the realized one-step Lyapunov increment by $\mathcal{J}_k \triangleq \mathcal{V}(\mathbf{z}_{k+1}) - \mathcal{V}(\mathbf{z}_k)$. Its conditional expectation

$$\mathcal{D}_k \triangleq \mathbb{E}[\mathcal{J}_k \mid \mathcal{F}_k]$$

is the Lyapunov drift at update k . The increment $\mathcal{J}_k$ is pathwise after the current queries are realized, whereas $\mathcal{D}_k$ averages it conditional on the pre-query history $\mathcal{F}_k$. Hence, $\mathcal{D}_k < 0$ represents expected one-step reduction of the combined stationarity and policy-performance measure, which makes its local minimization the step-size design objective.

Theorem 1 analyzes $+\mathbf{G}$ for the field energy ϕ , whereas Definition 1 extends the same geometry to $\mathcal{V}$ so that the local drift model can balance curvature use against policy performance.

Since $\mathcal{P} \geq 0$, $\mathbf{W}_V$ is positive definite, and $\lambda_F > 0$, Definition 1 gives $\mathcal{V} \geq 0$ and $\mathcal{V}(\mathbf{z}) = 0$ only if $\mathbf{z} \in \mathcal{Z}^*$. Under Assumption 1 and $\mathcal{P} \in C^3(\mathcal{D})$, the function $\mathcal{V}$ is C^3 with gradient $\nabla \mathcal{V} = (\lambda_F/\kappa_F) \nabla \mathcal{E}_F + (\lambda_P/\kappa_P) \nabla \mathcal{P}$, where $\nabla \mathcal{E}_F = \nabla \mathbf{F}^\top \mathbf{W}_V \mathbf{F}$. The weighted stationarity residual $\Phi(\mathbf{z}) \triangleq \mathbf{F}(\mathbf{z})^\top \mathbf{W}_V \mathbf{F}(\mathbf{z})$ satisfies $\Phi(\mathbf{z}) \leq 2\kappa_F \mathcal{V}(\mathbf{z})/\lambda_F$.

The composite Lyapunov decrease signals along the two

population directions are

$$\begin{aligned}D_F(\mathbf{z}) &\triangleq \langle \nabla \mathcal{V}(\mathbf{z}), \mathbf{F}(\mathbf{z}) \rangle \\ &= \frac{\lambda_F}{\kappa_F} \chi_F(\mathbf{z}) + \frac{\lambda_P}{\kappa_P} \langle \nabla \mathcal{P}(\mathbf{z}), \mathbf{F}(\mathbf{z}) \rangle, \\ D_G(\mathbf{z}) &\triangleq -\langle \nabla \mathcal{V}(\mathbf{z}), \mathbf{G}(\mathbf{z}) \rangle \\ &= \frac{\lambda_F}{\kappa_F} \chi_G(\mathbf{z}) - \frac{\lambda_P}{\kappa_P} \langle \nabla \mathcal{P}(\mathbf{z}), \mathbf{G}(\mathbf{z}) \rangle.\end{aligned}\quad (12)$$

Thus, $D_F > 0$ and $D_G > 0$ mean first-order decrease of $\mathcal{V}$ along $-\mathbf{F}$ and $+\mathbf{G}$, respectively.

B. Second-Order Local Drift Model

For a realized oracle pair and candidate step sizes (β, γ) , define the trial displacement $\Delta_k(\beta, \gamma) \triangleq -\beta \hat{\mathbf{F}}_k + \gamma \hat{\mathbf{G}}_k$ and the pathwise trial increment $\mathcal{J}_k(\beta, \gamma) \triangleq \mathcal{V}(\mathbf{z}_k + \Delta_k(\beta, \gamma)) - \mathcal{V}(\mathbf{z}_k)$. The actual quantities satisfy $\mathcal{J}_k = \mathcal{J}_k(\beta_k, \gamma_k)$ and $\mathcal{D}_k = \mathbb{E}[\mathcal{J}_k(\beta_k, \gamma_k) \mid \mathcal{F}_k]$. Exact query-dependent minimization of $\mathcal{D}_k$ requires the full conditional oracle distribution and repeated evaluation of the nonlinear post-update function $\mathcal{V}(\mathbf{z}_k + \Delta_k)$. A single query instead provides one realized oracle pair, so we minimize a second-order pathwise surrogate of $\mathcal{J}_k(\beta, \gamma)$ after observing $(\hat{\mathbf{F}}_k, \hat{\mathbf{G}}_k)$. The surrogate exposes the directional curvature along each search direction and their cross-curvature, which determine the step-size pair (β_k, γ_k) ; Section IV bounds its mismatch from the realized increment and converts the pathwise choice into an expected finite-time guarantee.

Assumption 3 (Local regularity along the iterates). *For every k and every feasible (β, γ) produced by the step-size routine, the segment $\{\mathbf{z}_k + t\Delta_k(\beta, \gamma) : t \in [0, 1]\} \subset \mathcal{D}$. There exist $\mathcal{F}_k$ -measurable $G_{V,k}, H_{V,k}, M_{3,k} \geq 0$ that, simultaneously for every feasible pair and almost every realization of the current stochastic queries, satisfy $\|\nabla \mathcal{V}(\mathbf{z}_k)\| \leq G_{V,k}$, $\|\nabla^2 \mathcal{V}(\mathbf{z}_k)\| \leq H_{V,k}$, and $\sup_{t \in [0,1]} \|\nabla^3 \mathcal{V}(\mathbf{z}_k + t\Delta_k(\beta, \gamma))\| \leq M_{3,k}$.*

Under Assumption 3, Taylor's theorem gives the following expansion for every feasible (β, γ) :

$$\begin{aligned}\mathcal{V}(\mathbf{z}_k + \Delta_k(\beta, \gamma)) - \mathcal{V}(\mathbf{z}_k) &= -\beta \langle \nabla \mathcal{V}(\mathbf{z}_k), \hat{\mathbf{F}}_k \rangle + \gamma \langle \nabla \mathcal{V}(\mathbf{z}_k), \hat{\mathbf{G}}_k \rangle \\ &\quad + \frac{1}{2} \beta^2 \hat{\mathbf{F}}_k^\top \nabla^2 \mathcal{V}(\mathbf{z}_k) \hat{\mathbf{F}}_k - \beta \gamma \hat{\mathbf{F}}_k^\top \nabla^2 \mathcal{V}(\mathbf{z}_k) \hat{\mathbf{G}}_k \\ &\quad + \frac{1}{2} \gamma^2 \hat{\mathbf{G}}_k^\top \nabla^2 \mathcal{V}(\mathbf{z}_k) \hat{\mathbf{G}}_k + \text{rem}_k(\beta, \gamma),\end{aligned}\quad (13)$$

The remainder satisfies $|\text{rem}_k(\beta, \gamma)| \leq \frac{M_{3,k}}{6} (\beta \|\hat{\mathbf{F}}_k\| + \gamma \|\hat{\mathbf{G}}_k\|)^3$.

Eq. (13) is the pathwise second-order Taylor expansion for each realized oracle pair. Its linear and quadratic terms define the five scalar coefficients

$$\begin{aligned}e_k &\triangleq \langle \nabla \mathcal{V}(\mathbf{z}_k), \hat{\mathbf{F}}_k \rangle, & d_k &\triangleq -\langle \nabla \mathcal{V}(\mathbf{z}_k), \hat{\mathbf{G}}_k \rangle, \\ c_k &\triangleq \hat{\mathbf{F}}_k^\top \nabla^2 \mathcal{V}(\mathbf{z}_k) \hat{\mathbf{F}}_k, & a_k &\triangleq \hat{\mathbf{G}}_k^\top \nabla^2 \mathcal{V}(\mathbf{z}_k) \hat{\mathbf{G}}_k, \\ b_k &\triangleq \hat{\mathbf{F}}_k^\top \nabla^2 \mathcal{V}(\mathbf{z}_k) \hat{\mathbf{G}}_k.\end{aligned}\quad (14)$$

Eq. (14) is pathwise: it uses the realized oracle pair $(\hat{\mathbf{F}}_k, \hat{\mathbf{G}}_k)$ rather than the population directions $(\mathbf{F}(\mathbf{z}_k), \mathbf{G}(\mathbf{z}_k))$. Collecting these five coefficients gives the pathwise reference quadratic

$$q_k(\beta, \gamma) \triangleq -\beta e_k - \gamma d_k + \frac{1}{2} c_k \beta^2 - b_k \beta \gamma + \frac{1}{2} a_k \gamma^2. \quad (15)$$

Thus, Eq. (13) is $\mathcal{J}_k(\beta, \gamma) = q_k(\beta, \gamma) + \text{rem}_k(\beta, \gamma)$. Here e_k, d_k are first-order predicted decreases along $-\hat{\mathbf{F}}_k, \hat{\mathbf{G}}_k$, c_k, a_k are their directional curvatures, and b_k is the cross-curvature. Assumptions 3 and 2 respectively control the third-order remainder and the conditional oracle errors used in the expected-drift analysis.

C. Closed-Form Box-QP Step-Size Selection

Discarding the bounded third-order remainder in Eq. (13) yields a bivariate quadratic model for selecting (β, γ) . Eq. (14) defines reference coefficients from $\nabla \mathcal{V}(\mathbf{z}_k)$ and directional Hessian actions. We denote their evaluated counterparts returned by the local-model routine by $(\tilde{e}_k, \tilde{d}_k, \tilde{c}_k, \tilde{a}_k, \tilde{b}_k)$. Exact automatic differentiation gives $(\tilde{e}_k, \tilde{d}_k, \tilde{c}_k, \tilde{a}_k, \tilde{b}_k) = (e_k, d_k, c_k, a_k, b_k)$, whereas finite-difference or directional-secant implementations, possibly combined with moving averages, generally give estimates such as $\tilde{e}_k \approx e_k$ and $\tilde{c}_k \approx c_k$. The quadratic coefficient matrix assembled from the evaluated values is $\hat{\mathbf{H}}_k \triangleq \begin{bmatrix} \tilde{c}_k & -\tilde{b}_k \\ -\tilde{b}_k & \tilde{a}_k \end{bmatrix}$. Because local nonconvexity or evaluation error may make $\hat{\mathbf{H}}_k$ indefinite or nearly singular, we stabilize it by a common spectral shift. Let $\lambda_{\min}(\cdot)$ denote the smallest eigenvalue and fix a positive-definiteness floor $\lambda_{\text{pd}} > 0$. Using $[u]_+ \triangleq \max\{u, 0\}$ for the positive part of a scalar u , the shift is

$$\lambda_k = [\lambda_{\text{pd}} - \lambda_{\min}(\hat{\mathbf{H}}_k)]_+, \quad \hat{\mathbf{H}}_k = \tilde{\mathbf{H}}_k + \lambda_k \mathbf{I}_2. \quad (16)$$

Here λ_k is the smallest nonnegative common diagonal correction that raises the minimum eigenvalue to at least λ_{pd} . We denote the resulting numerically stabilized coefficients by hats: $\hat{e}_k = \tilde{e}_k, \hat{d}_k = \tilde{d}_k, \hat{b}_k = \tilde{b}_k$, and $(\hat{c}_k, \hat{a}_k) = (\tilde{c}_k + \lambda_k, \tilde{a}_k + \lambda_k)$. Equivalently, $\rho_{c,k} = \rho_{a,k} = \lambda_k \geq 0$ add $\frac{1}{2}\rho_{c,k}\beta^2 + \frac{1}{2}\rho_{a,k}\gamma^2$ to the tilde-coefficient model while leaving its linear and cross terms unchanged. The implemented model is therefore

$$\hat{q}_k(\beta, \gamma) \triangleq -\beta\hat{e}_k - \gamma\hat{d}_k + \frac{1}{2}\hat{c}_k\beta^2 - \hat{b}_k\beta\gamma + \frac{1}{2}\hat{a}_k\gamma^2, \quad (\beta, \gamma) \in [0, \infty)^2. \quad (17)$$

Its stabilized Hessian is $\hat{\mathbf{H}}_k$, and $\hat{\delta}_k \triangleq \det \hat{\mathbf{H}}_k = \hat{c}_k\hat{a}_k - \hat{b}_k^2$.

Eq. (16) gives $\lambda_{\min}(\hat{\mathbf{H}}_k) = \max\{\lambda_{\min}(\tilde{\mathbf{H}}_k), \lambda_{\text{pd}}\} \geq \lambda_{\text{pd}}$. Thus, $\hat{\mathbf{H}}_k$ is positive definite, and its leading principal minors satisfy $\hat{c}_k > 0$ and $\hat{\delta}_k > 0$. The controller thus solves the strictly convex two-variable quadratic program $\min_{(\beta, \gamma) \in [0, \infty)^2} \hat{q}_k(\beta, \gamma)$. Its 2×2 Hessian permits an explicit interior solution together with a finite set of boundary candidates. The unconstrained critical point is

$$(\beta_k^{\text{uc}}, \gamma_k^{\text{uc}}) \triangleq \left(\frac{\hat{a}_k\hat{e}_k + \hat{b}_k\hat{d}_k}{\hat{\delta}_k}, \frac{\hat{b}_k\hat{e}_k + \hat{c}_k\hat{d}_k}{\hat{\delta}_k} \right), \quad (18)$$

and the boundary candidates are $([\hat{e}_k/\hat{c}_k]_+, 0)$, $(0, [\hat{d}_k/\hat{a}_k]_+)$, $(0, 0)$.

Proposition 2 (Closed-form two-direction surrogate controller). *The implemented problem in Eq. (17) has a unique minimizer. For the uncapped problem, it is given by Eq. (18) when both components are nonnegative and otherwise by the boundary candidate with smallest $\hat{q}_k$. When the unconstrained critical point belongs to $[0, \infty)^2$, the predicted decrease $\hat{\Delta}_k^* \triangleq -\hat{q}_k(\beta_k^*, \gamma_k^*)$ is*

$$\hat{\Delta}_k^* = \frac{\hat{a}_k\hat{e}_k^2 + 2\hat{b}_k\hat{e}_k\hat{d}_k + \hat{c}_k\hat{d}_k^2}{2\hat{\delta}_k}. \quad (19)$$

For finite caps $\beta_{\max}, \gamma_{\max} > 0$, the unique minimizer on $[0, \beta_{\max}] \times [0, \gamma_{\max}]$ is the member with smallest objective value among the feasible unconstrained point and the four edge minimizers

$$\begin{aligned} &([\hat{e}_k/\hat{c}_k]_{[0, \beta_{\max}]}, 0), \quad (0, [\hat{d}_k/\hat{a}_k]_{[0, \gamma_{\max}]}) \\ &(\beta_{\max}, [\hat{d}_k + \hat{b}_k\beta_{\max}]/\hat{a}_k]_{[0, \gamma_{\max}]}), \\ &([\hat{e}_k + \hat{b}_k\gamma_{\max}]/\hat{c}_k]_{[0, \beta_{\max}]}, \gamma_{\max}), \end{aligned}$$

where $[u]_{[l, r]} \triangleq \min\{r, \max\{l, u\}\}$. Consequently, its predicted decrease is at least as large as that of every single-direction or fixed-coupling comparator whose step-size pair lies in the same box.

The proposition provides both the analytical uncapped rule and the exact safeguarded box solution used by the implementation.

D. Model-Level Comparison and Algorithm Summary

The closed-form rule makes the two-direction update computationally explicit. Its model-level benefit is determined by comparing the optimized quadratic decrease with the decreases attainable after restricting the feasible step-size pair.

Proposition 3 (Predicted-decrease comparison). *Fix a maximum comparator step size $s_{\max} > 0$. Let $\hat{\Delta}_k^{\mathbf{F}}$ and $\hat{\Delta}_k^{\mathbf{G}}$ denote the best predicted decreases along the two coordinate axes, and let $\hat{\Delta}_k^{2\text{nd}}$ denote the best decrease along the PPM/EGM coupling $(\beta, \gamma) = (s, s^2)$:*

$$\begin{aligned} \hat{\Delta}_k^{\mathbf{F}} &= \frac{[\hat{e}_k]_+^2}{2\hat{c}_k}, \quad \hat{\Delta}_k^{\mathbf{G}} = \frac{[\hat{d}_k]_+^2}{2\hat{a}_k}, \\ \hat{\Delta}_k^{2\text{nd}} &= - \min_{s \in [0, s_{\max}]} \hat{q}_k(s, s^2). \end{aligned}$$

Then $\hat{\Delta}_k^ \geq \max\{\hat{\Delta}_k^{\mathbf{F}}, \hat{\Delta}_k^{\mathbf{G}}, \hat{\Delta}_k^{2\text{nd}}\}$ almost surely. Moreover, when the unconstrained critical point in Eq. (18) belongs to $[0, \infty)^2$, the gap over the field-only rule is exactly*

$$\hat{\Delta}_k^* - \hat{\Delta}_k^{\mathbf{F}} = \frac{(\hat{b}_k\hat{e}_k + \hat{c}_k\hat{d}_k)^2}{2\hat{c}_k\hat{\delta}_k} + \frac{[-\hat{e}_k]_+^2}{2\hat{c}_k} \geq 0, \quad (20)$$

which is strictly positive if and only if either $\hat{b}_k\hat{e}_k + \hat{c}_k\hat{d}_k \neq 0$ or $\hat{e}_k < 0$. In the usual field-descent regime $\hat{e}_k \geq 0$, strict improvement is therefore equivalent to the curvature direction having a nonzero reduced gradient after projecting out the field direction in the $\hat{\mathbf{H}}_k$ -metric.

Eq. (20) measures the curvature direction's additional predicted decrease through the reduced gradient $\hat{b}_k\hat{e}_k + \hat{c}_k\hat{d}_k$ left after accounting for the field direction. For collinear directions, the common spectral shift makes the Hessian nonsingular; because the direction span is unchanged, any gain arises from step-size constraints or regularization. One joint policy update of the resulting controller consists of the following operations:

- (i) obtain the stochastic field and curvature observations $(\hat{\mathbf{F}}_k, \hat{\mathbf{G}}_k)$;
- (ii) evaluate the five coefficients of the local Lyapunov model and apply the common spectral shift;

(iii) solve the two-dimensional box QP for (β_k, γ_k) using Proposition 2;

(iv) apply the stochastic policy recursion in Eq. (9).

This sequence separates the geometric construction of the two search directions from their Lyapunov-based step-size selection. Section IV combines this comparator dominance with the field-geometry certificate and explicitly propagates stochastic-oracle, model-estimation, Taylor-remainder, and safeguard errors into finite-time guarantees.

IV. FINITE-TIME AND POLICY-SPACE GUARANTEES

Section III constructs the controller and characterizes its predicted quadratic decrease. We now bound its realized and conditional Lyapunov changes, derive a finite-time certificate for the two available descent directions, and then identify when this certificate yields stationarity and policy-space performance guarantees.

A. Conditional Drift and a Two-Direction Certificate

Assumption 4 (Local admissible region). *There exist an open set $\mathcal{N}$ whose compact closure satisfies $\bar{\mathcal{N}} \subset \mathcal{D}$ and an index $k_0 \geq 0$ such that $\mathcal{N}$ contains the local stationary component of interest, $\mathcal{P}$ is C^3 on a neighborhood of $\bar{\mathcal{N}}$, and $\mathbf{z}_k \in \mathcal{N}$ for every $k \geq k_0$ almost surely.*

This is a local-invariance condition: the safeguards in Section V keep the accepted trajectory in a prescribed implementable region, and the analysis applies when that trajectory remains in $\mathcal{N}$. Because $\bar{\mathcal{N}}$ is compactly contained in $\mathcal{D}$, continuity yields uniform constants $G_V, H_V, B_F, B_G \geq 0$ such that $\|\nabla \mathcal{V}\| \leq G_V$, $\|\nabla^2 \mathcal{V}\| \leq H_V$, $\|\mathbf{F}\| \leq B_F$, and $\|\mathbf{G}\| \leq B_G$ on $\mathcal{N}$. In particular, $G_{V,k}, H_{V,k}$ may be replaced by G_V, H_V for $k \geq k_0$.

Recall the pathwise reference quadratic q_k in Eq. (15) and the stabilized implemented quadratic $\hat{q}_k$ in Eq. (17). For fixed caps $\beta_{\max}, \gamma_{\max} > 0$ and an implemented pair $(\beta_k, \gamma_k) \in [0, \beta_{\max}] \times [0, \gamma_{\max}]$, define the box-optimal predicted decrease

$$\hat{\Delta}_k^{\text{box}} \triangleq - \min_{0 \leq \beta \leq \beta_{\max}, 0 \leq \gamma \leq \gamma_{\max}} \hat{q}_k(\beta, \gamma).$$

The implemented model $\hat{q}_k$ uses the stabilized coefficients, and one-sided residuals measure model discrepancy and accepted step-size-pair suboptimality. Define the cubic Taylor remainder $\varepsilon_k^{\text{rem}} \triangleq M_{3,k}(\beta_k \|\hat{\mathbf{F}}_k\| + \gamma_k \|\hat{\mathbf{G}}_k\|)^3/6$. Let $\varepsilon_k^{\text{model}} \geq 0$ be $\mathcal{F}_{k+1}$ -measurable and satisfy $q_k(\beta_k, \gamma_k) \leq \hat{q}_k(\beta_k, \gamma_k) + \varepsilon_k^{\text{model}}$. For the implemented box problem, define $\varepsilon_k^{\text{step,box}} \triangleq \hat{q}_k(\beta_k, \gamma_k) + \hat{\Delta}_k^{\text{box}} \geq 0$ and

$$\varepsilon_k^{\text{box}} \triangleq \varepsilon_k^{\text{rem}} + \varepsilon_k^{\text{model}} + \varepsilon_k^{\text{step,box}}. \quad (21)$$

Suppose the raw evaluated coefficients satisfy $|\tilde{e}_k - e_k| \leq \delta_{e,k}$, $|\tilde{d}_k - d_k| \leq \delta_{d,k}$, $|\tilde{c}_k - c_k| \leq \delta_{c,k}$, $|\tilde{a}_k - a_k| \leq \delta_{a,k}$, and $|\tilde{b}_k - b_k| \leq \delta_{b,k}$ for nonnegative error bounds.

Proposition 4 (Local-model and pathwise error bounds). *Under the preceding coefficient-error bounds, the following statements hold.*

(i) *For every feasible (β, γ) ,*

$$\begin{aligned} \varepsilon_k^{\text{model}}(\beta, \gamma) &\triangleq \beta \delta_{e,k} + \gamma \delta_{d,k} + \frac{1}{2} \beta^2 \delta_{c,k} \\ &\quad + \beta \gamma \delta_{b,k} + \frac{1}{2} \gamma^2 \delta_{a,k}, \\ \varepsilon_k^{\text{cmp}}(\beta, \gamma) &\triangleq \varepsilon_k^{\text{model}}(\beta, \gamma) + \frac{1}{2} \rho_{c,k} \beta^2 + \frac{1}{2} \rho_{a,k} \gamma^2, \\ q_k - \hat{q}_k &\leq \varepsilon_k^{\text{model}}, \\ \hat{q}_k - q_k &\leq \varepsilon_k^{\text{cmp}}. \end{aligned} \quad (22)$$

The inequality arguments are suppressed, cmp denotes the comparator-side bound, and Eq. (21) uses $\varepsilon_k^{\text{model}} = \varepsilon_k^{\text{model}}(\beta_k, \gamma_k)$.

(ii) *Under Assumption 3, the realized update satisfies*

$$\mathcal{V}(\mathbf{z}_{k+1}) - \mathcal{V}(\mathbf{z}_k) \leq -\hat{\Delta}_k^{\text{box}} + \varepsilon_k^{\text{box}} \quad \text{almost surely.} \quad (23)$$

(iii) *If Assumptions 2 and 3 hold and $\beta_k \leq \beta_{\max}$, $\gamma_k \leq \gamma_{\max}$, then*

$$\mathbb{E}[\varepsilon_k^{\text{rem}} | \mathcal{F}_k] \leq \frac{2M_{3,k}}{3} (\beta_{\max} + \gamma_{\max})^3 \sigma_{3,k}^3. \quad (24)$$

If $M_{3,k} \leq \bar{M}_3$ and $\sigma_{3,k} \leq \bar{\sigma}_3$, then $\mathbb{E}[\varepsilon_k^{\text{rem}} | \mathcal{F}_k] \leq \bar{R}_{\text{rem}}$ almost surely, where $\bar{R}_{\text{rem}} \triangleq (2\bar{M}_3/3)(\beta_{\max} + \gamma_{\max})^3 \bar{\sigma}_3^3$.

The three parts control coefficient evaluation, pathwise model mismatch, and the conditional Taylor remainder; exact automatic differentiation and an exact box-QP solution set the corresponding evaluation, selected-pair, and solution residuals to zero.

Assumption 5 (Uniform local budgets). *There exist finite nonnegative constants appearing below such that, for every $k \geq k_0$, the following hold: (a) $\bar{b}_k^F \leq \bar{b}_F$, $\bar{b}_k^G \leq \bar{b}_G$, $\sigma_{2,k} \leq \bar{\sigma}_2$, and $\sigma_{3,k} \leq \bar{\sigma}_3$; (b) the common spectral shift satisfies $\rho_{c,k} = \rho_{a,k} \leq \bar{\rho}$, and the raw coefficient-evaluation errors in Proposition 4 satisfy*

$$\begin{aligned} \mathbb{E}[\delta_{e,k} | \mathcal{F}_k] &\leq \bar{\delta}_e, \quad \mathbb{E}[\delta_{d,k} | \mathcal{F}_k] \leq \bar{\delta}_d, \\ \mathbb{E}[\delta_{c,k} | \mathcal{F}_k] &\leq \bar{\delta}_c, \quad \mathbb{E}[\delta_{a,k} | \mathcal{F}_k] \leq \bar{\delta}_a, \quad \mathbb{E}[\delta_{b,k} | \mathcal{F}_k] \leq \bar{\delta}_b; \\ (c) \quad M_{3,k} &\leq \bar{M}_3 \text{ and } \mathbb{E}[\varepsilon_k^{\text{step,box}} | \mathcal{F}_k] \leq \bar{R}_{\text{step}} \text{ almost surely.} \end{aligned}$$

Define the derived oracle-moment bounds and aggregate residual budgets

$$\begin{aligned} M_F &\triangleq 3\bar{B}_F^2 + 3\bar{b}_F^2 + 3\bar{\sigma}_2^2, \quad M_G \triangleq 3\bar{B}_G^2 + 3\bar{b}_G^2 + 3\bar{\sigma}_2^2, \\ \bar{b}_D &\triangleq \max\{\bar{b}_F, \bar{b}_G\}, \quad M_D \triangleq \max\{M_F, M_G\}, \\ \bar{\delta}_1 &\triangleq \max\{\bar{\delta}_e, \bar{\delta}_d\}, \quad \bar{\delta}_2 \triangleq \max\{\bar{\delta}_c, \bar{\delta}_a\}, \\ \bar{R}_{\text{model}} &\triangleq \beta_{\max} \bar{\delta}_e + \gamma_{\max} \bar{\delta}_d + \frac{1}{2} \beta_{\max}^2 \bar{\delta}_c \\ &\quad + \beta_{\max} \gamma_{\max} \bar{\delta}_b + \frac{1}{2} \gamma_{\max}^2 \bar{\delta}_a, \\ \bar{R}_{\text{box}} &\triangleq \bar{R}_{\text{rem}} + \bar{R}_{\text{model}} + \bar{R}_{\text{step}}. \end{aligned}$$

Proposition 4, the step-size caps, and Assumption 5 give $\mathbb{E}[\varepsilon_k^{\text{box}} | \mathcal{F}_k] \leq \bar{R}_{\text{box}}$ almost surely, so the aggregate is derived from its three displayed sources. For a comparator step size $s \in [0, \min\{\beta_{\max}, \gamma_{\max}\}]$, let

$$\mathcal{R}_D(s) \triangleq s(G_V \bar{b}_D + \bar{\delta}_1) + \frac{1}{2} s^2 (H_V M_D + \bar{\rho} + \bar{\delta}_2) + \bar{R}_{\text{box}}. \quad (25)$$

Proposition 5 (Conditional and finite-time two-direction Lyapunov certificate). *Suppose Assumptions 1, 2, 3, 4, and 5 hold. Then the following statements hold.*

(i) For every admissible deterministic comparator step size s ,

$$\mathcal{D}_k \leq -s [\max\{D_F(\mathbf{z}_k), D_G(\mathbf{z}_k)\}]_+ + \mathcal{R}_D(s). \quad (26)$$

(ii) For every $s \in (0, \min\{\beta_{\max}, \gamma_{\max}\}]$ and integer $K > k_0$,

$$\begin{aligned} & \frac{1}{K - k_0} \sum_{k=k_0}^{K-1} \mathbb{E}[\max\{D_F(\mathbf{z}_k), D_G(\mathbf{z}_k)\}]_+ \\ & \leq \frac{\mathbb{E}[\mathcal{V}(\mathbf{z}_{k_0})]}{s(K - k_0)} + \frac{\mathcal{R}_D(s)}{s}. \end{aligned} \quad (27)$$

Part (i) transfers the strongest available single-direction decrease to the implemented update up to explicit residuals, and Part (ii) converts this one-step comparison into a finite-time average certificate. The next subsection relates this signal to the stationarity residual.

B. From Geometric Coverage to Finite-Time Stationarity

For this conversion, define the geometric coverage modulus on $\mathcal{N}$ by

$$m_{\text{geo}} \triangleq \inf_{\substack{\mathbf{z} \in \mathcal{N} \\ \Phi(\mathbf{z}) > 0}} \frac{\max\{\chi_F(\mathbf{z}), \chi_G(\mathbf{z})\}}{\Phi(\mathbf{z})}. \quad (28)$$

The condition $m_{\text{geo}} > 0$ states that, at every nonstationary point in $\mathcal{N}$, at least one available direction supplies field-energy decrease proportional to the stationarity residual. The following proposition derives this property locally from the Jacobian at the target stationary point; $\sigma_{\min}(\cdot)$ denotes the smallest singular value.

Proposition 6 (A sufficient condition for geometric coverage). *Fix a target stationary point $\mathbf{z}^* \in \mathcal{Z}^*$ and define its metric-transformed Jacobian $\tilde{\mathbf{A}}_* \triangleq \mathbf{W}_V^{1/2} \mathbf{A}(\mathbf{z}^*) \mathbf{W}_V^{-1/2}$. If $\text{sym}(\tilde{\mathbf{A}}_*)$ is positive semidefinite and $\sigma_{\min}(\tilde{\mathbf{A}}_*) > 0$, then there exist a constant $c_* > 0$ and an open neighborhood $\mathcal{N}_*$ of $\mathbf{z}^*$, with $\mathcal{N}_* \subset \mathcal{D}$, such that $\max\{\chi_F(\mathbf{z}), \chi_G(\mathbf{z})\} \geq c_* \Phi(\mathbf{z})$ for every $\mathbf{z} \in \mathcal{N}_*$. Consequently, if the admissible region in Assumption 4 satisfies $\bar{\mathcal{N}} \subset \mathcal{N}_*$, then $m_{\text{geo}} \geq c_* > 0$.*

For $\mathbf{W}_V = \mathbf{I}$, max-variable concavity and min-variable convexity at $\mathbf{z}^*$ give the positive-semidefinite condition, while nonsingularity gives a nondegenerate target linearization; continuity then supplies neighborhood coverage, including pure rotation with $\text{sym}(\tilde{\mathbf{A}}_*) = 0$. For the normal block $\mathbf{A} = \mu \mathbf{I}_2 + \sigma \mathbf{J}_0$, the exact ratio in Eq. (28) is $\max\{\mu, \sigma^2 - \mu^2\}$; in particular, $\mu = 0$ gives $m_{\text{geo}} = \sigma^2$.

Compactness and smoothness give the finite performance-direction bound

$$B_{P,D} \triangleq \sup_{\mathbf{z} \in \bar{\mathcal{N}}} \max\{|\langle \nabla \mathcal{P}, \mathbf{F} \rangle|, |\langle \nabla \mathcal{P}, \mathbf{G} \rangle|\}.$$

When Assumption 4 holds and $m_{\text{geo}} > 0$, define the geometric descent coefficient μ_{geo} and policy-component perturbation r_P by

$$\mu_{\text{geo}} \triangleq \frac{\lambda_F m_{\text{geo}}}{\kappa_F}, \quad r_P \triangleq \frac{\lambda_P B_{P,D}}{\kappa_P}.$$

Selecting the larger signal in Eq. (12) and applying Eq. (28) gives, for every $\mathbf{z} \in \mathcal{N}$,

$$\max\{D_F(\mathbf{z}), D_G(\mathbf{z})\} \geq \mu_{\text{geo}} \Phi(\mathbf{z}) - r_P. \quad (29)$$

Proposition 6 and Eq. (29) provide the stationarity conversion: target-Jacobian conditions yield local coverage, which lower-bounds the strongest two-direction Lyapunov signal by the stationarity residual up to r_P . The perturbation r_P vanishes for $\lambda_P = 0$ and otherwise enters the residual floor.

Theorem 2 (Main Theorem: Local Finite-Time Stationarity under Geometric Coverage). *Suppose Assumptions 1, 2, 3, 4, and 5 hold, and the modulus in Eq. (28) satisfies $m_{\text{geo}} > 0$. For any comparator step size $s \in (0, \min\{\beta_{\max}, \gamma_{\max}\}]$, define $\mathcal{R}_{\text{geo}}(s) \triangleq \mathcal{R}_D(s) + sr_P$. Then, for every integer $K > k_0$,*

$$\frac{1}{K - k_0} \sum_{k=k_0}^{K-1} \mathbb{E}[\Phi(\mathbf{z}_k)] \leq \frac{\mathbb{E}[\mathcal{V}(\mathbf{z}_{k_0})]}{s\mu_{\text{geo}}(K - k_0)} + \frac{\mathcal{R}_{\text{geo}}(s)}{s\mu_{\text{geo}}}. \quad (30)$$

For fixed admissible s and K -independent budgets, Eq. (30) gives an $O(1/K)$ transient to an explicit floor containing the oracle, model, Taylor-remainder, policy-component, and safeguard errors. The signals χ_F and χ_G respectively certify contractive symmetric fields and pure rotation, and box-QP dominance inherits either certificate. For an independent uniform index $I_K \in \{k_0, \dots, K-1\}$, the left side is $\mathbb{E}[\Phi(\mathbf{z}_{I_K})]$, which gives an equivalent uniform-random-iterate interpretation of the theorem. The comparator s trades nominal decrease against higher-order and s -independent residuals, while the implemented controller continues to select (β_k, γ_k) from the box QP.

C. Policy-Space Performance Guarantee

Exploitability and robust return are policy-space quantities beyond parameter-space stationarity. We therefore transfer Lyapunov control to these performance measures. For stationary randomized policies, define the maximizing player's worst-case return, game value, and unregularized Nash gap (reported below as exploitability), where the subscript BR denotes best response:

$$\begin{aligned} R_{\text{BR}}(\pi) & \triangleq \inf_{\bar{\nu}} J(\pi, \bar{\nu}), \\ v^* & \triangleq \sup_{\bar{\pi}} R_{\text{BR}}(\bar{\pi}), \\ \text{Gap}_0(\pi, \nu) & \triangleq \sup_{\bar{\pi}} J(\bar{\pi}, \nu) - \inf_{\bar{\nu}} J(\pi, \bar{\nu}). \end{aligned} \quad (31)$$

The following proposition links the entropy-regularized gap used by the controller to the original zero-sum Markov-game objective.

Proposition 7 (Policy-space performance bridge). *Suppose $\mathcal{A}$ and $\mathcal{B}$ are finite. For every policy pair (π, ν) and $\varepsilon \geq 0$,*

$$\begin{aligned} 0 \leq v^* - R_{\text{BR}}(\pi) & \leq \text{Gap}_0(\pi, \nu) \\ & \leq \text{Gap}_{\varepsilon}(\pi, \nu) + \frac{\varepsilon(\log |\mathcal{A}| + \log |\mathcal{B}|)}{1 - \alpha}. \end{aligned} \quad (32)$$

Proof. The definitions in Eq. (31) give the first inequality, while $v^* \leq \sup_{\bar{\pi}} J(\bar{\pi}, \nu)$ gives the second. The discounted entropy bounds are $0 \leq H_{\alpha}^{\pi} \leq \log |\mathcal{A}|/(1 - \alpha)$ and $0 \leq H_{\alpha}^{\nu} \leq \log |\mathcal{B}|/(1 - \alpha)$. Substituting them into the two best-response terms in Eq. (31) gives the final inequality in Eq. (32). $\square$

Thus the regularized gap controls the maximizing player's unregularized worst-case-return loss up to the stated entropy term.

Proposition 8 (Smooth finite-game performance component). *Suppose $\mathcal{S}, \mathcal{A}, \mathcal{B}$ are finite, the one-stage rewards are bounded, $\varepsilon > 0$, and, on the parameter neighborhood under consideration, the current policies are C^3 with action probabilities bounded away from zero. Then the entropy-regularized best-response values against either current player, and hence Gap_ε , are C^3 functions of the current policy parameters.*

Proof. For a fixed opponent policy, the finite-state soft Bellman operator is C^3 in its value vector and the opponent parameters because it is composed of smooth policy probabilities, finite sums, and log-sum-exp. Its value-vector Jacobian is a discounted transition matrix and therefore has induced infinity norm at most $\alpha < 1$. The implicit-function theorem consequently makes its unique fixed point and regularized best-response value C^3 in the opponent parameters. Applying the same argument to the soft-min Bellman operator makes both best-response values, and hence their difference Gap_ε , C^3 . $\square$

Consequently, choosing $\mathcal{P} = \text{Gap}_\varepsilon$ gives a concrete smooth performance component for Definition 1 in finite Markov games.

Assumption 6 (Local policy-space error bound). *There exist constants $c_{\text{gap}} > 0$ and $\xi_{\text{gap}} \geq 0$ such that*

$$\text{Gap}_\varepsilon(\mathbf{z}) \leq c_{\text{gap}} \Phi(\mathbf{z}) + \xi_{\text{gap}}, \quad \mathbf{z} \in \mathcal{N}. \quad (33)$$

Eq. (33) is a local Polyak–Łojasiewicz-type policy error bound [30]. The slack ξ_{gap} accommodates parameter-stationary points with nonzero policy-space exploitability and hence allows the two zero sets to differ.

Corollary 1 (Ergodic policy-space performance). *Suppose the assumptions of Theorem 2 and Assumption 6 hold, and $\mathcal{A}$ and $\mathcal{B}$ are finite. Then*

$$\begin{aligned} & \frac{1}{K - k_0} \sum_{k=k_0}^{K-1} \mathbb{E}[v^* - R_{\text{BR}}(\pi_{\theta_k})] \\ & \leq \frac{1}{K - k_0} \sum_{k=k_0}^{K-1} \mathbb{E}[\text{Gap}_0(\pi_{\theta_k}, \nu_{\psi_k})] \\ & \leq c_{\text{gap}} \left[\frac{\mathbb{E}[\mathcal{V}(\mathbf{z}_{k_0})]}{s\mu_{\text{geo}}(K - k_0)} + \frac{\mathcal{R}_{\text{geo}}(s)}{s\mu_{\text{geo}}} \right] + \xi_{\text{gap}} + B_\varepsilon, \end{aligned} \quad (34)$$

where $B_\varepsilon \triangleq \varepsilon(\log |\mathcal{A}| + \log |\mathcal{B}|)/(1 - \alpha)$ is the entropy-regularization transfer term.

Proof. Average Eq. (33), apply Theorem 2, and then use Proposition 7 pointwise. $\square$

Thus, Eq. (34) gives the policy-space consequence of the main stationarity theorem under the local error bound with slack.

V. PRACTICAL IMPLEMENTATION

This section specifies how the local quantities required by the controller can be evaluated in practice. We consider exact automatic differentiation, directional-secant approximations, and safeguarded box-constrained updates. For finite games, the implementation evaluates both the saddle field and the regularized gap from J_ε with $\varepsilon > 0$; the unregularized return J is retained for the performance measures in Section IV-C.

A. Exact local-model evaluation

The reference pathwise terms in Eq. (14) are assembled from one field oracle $\hat{\mathbf{F}}_k$, one curvature oracle $\hat{\mathbf{G}}_k$, one gradient $\nabla \mathcal{V}(\mathbf{z}_k)$, and the Hessian–vector products of $\mathcal{V}$ along $\hat{\mathbf{F}}_k$ and $\hat{\mathbf{G}}_k$. Automatic differentiation can pass through a fixed differentiable inner loop for $\mathcal{P}$, while implicit differentiation handles an exact local envelope. Because $\mathcal{E}_F$ contains $\mathbf{F}^\top \mathbf{W}_V \mathbf{F}/2$, exact Hessian–vector products of $\mathcal{V}$ generally involve third derivatives of the game objective; the secant realization below avoids forming them. Exact automatic differentiation gives $(\tilde{e}_k, \tilde{d}_k, \tilde{c}_k, \tilde{a}_k, \tilde{b}_k) = (e_k, d_k, c_k, a_k, b_k)$. After exact coefficient evaluation, $\tilde{q}_k - q_k = \lambda_k(\beta^2 + \gamma^2)/2 \geq 0$; hence the selected-pair one-sided model residual is zero, while the comparator-side bound retains the explicit shift.

B. Approximate local-model estimation

For probe radii $\delta_{F,k}, \delta_{G,k} > 0$, a directional-secant realization holds $(\hat{\mathbf{F}}_k, \hat{\mathbf{G}}_k)$ fixed and evaluates $V_{\pm F} = \mathcal{V}(\mathbf{z}_k \pm \delta_{F,k} \hat{\mathbf{F}}_k)$, $V_{\pm G} = \mathcal{V}(\mathbf{z}_k \pm \delta_{G,k} \hat{\mathbf{G}}_k)$, and $V_{\pm F, \pm G} = \mathcal{V}(\mathbf{z}_k \pm \delta_{F,k} \hat{\mathbf{F}}_k \pm \delta_{G,k} \hat{\mathbf{G}}_k)$. Writing $V_0 = \mathcal{V}(\mathbf{z}_k)$, the central stencils give the tilde approximations

$$\begin{aligned} \tilde{e}_k &= \frac{V_{+F} - V_{-F}}{2\delta_{F,k}}, & \tilde{d}_k &= -\frac{V_{+G} - V_{-G}}{2\delta_{G,k}}, \\ \tilde{c}_k &= \frac{V_{+F} - 2V_0 + V_{-F}}{\delta_{F,k}^2}, & \tilde{a}_k &= \frac{V_{+G} - 2V_0 + V_{-G}}{\delta_{G,k}^2}, \\ \tilde{b}_k &= \frac{V_{+F,+G} - V_{+F,-G} - V_{-F,+G} + V_{-F,-G}}{4\delta_{F,k}\delta_{G,k}}, \end{aligned}$$

before the spectral shift in Section III-C is applied. Writing $\hat{\mathbf{F}}(\mathbf{z}, \zeta)$ for the field oracle evaluated at $\mathbf{z}$ with random seed ζ , a shared seed ζ_k and radius $\epsilon_k > 0$ give the Jacobian–vector finite-difference curvature oracle $\hat{\mathbf{G}}_k^{\text{fd}} = [\hat{\mathbf{F}}(\mathbf{z}_k + \epsilon_k \hat{\mathbf{F}}_k, \zeta_k) - \hat{\mathbf{F}}(\mathbf{z}_k, \zeta_k)]/\epsilon_k$. A moment-based realization smooths the second-order terms by exponential moving averages while preserving the subproblem.

The effect of these approximation errors on the one-step model is quantified by Proposition 4.

C. Safeguarded computation

With positive caps β_{max} and γ_{max} , the implementation solves $\min_{(\beta, \gamma) \in [0, \beta_{\text{max}}] \times [0, \gamma_{\text{max}}]} \hat{q}_k(\beta, \gamma)$. Any loss from approximate solution or backtracking relative to the box optimum is recorded in $\varepsilon_k^{\text{step, box}}$ in Theorem 2. Each iteration queries $(\hat{\mathbf{F}}_k, \hat{\mathbf{G}}_k)$, evaluates the five tilde coefficients, applies the spectral shift, solves the two-dimensional box QP by Proposition 2, and applies Eq. (9). If the trial segment leaves a prescribed implementable trust region $\mathcal{T} \subset \mathcal{D}$, a common backtracking factor shrinks (β_k, γ_k) ; the analysis assumes the accepted trajectory lies in $\mathcal{N}$, and records lost model decrease in $\varepsilon_k^{\text{step, box}}$. For bounded $\mathcal{T}$, the implementation reduces the probe radii as needed until every directional-stencil point lies in $\mathcal{T}$.

VI. EXPERIMENTS

The experiments follow the same hierarchy as the analysis. The normal-form study first isolates the symmetric–antisymmetric mechanism, the tabular and neural population-oracle studies then test policy-space behavior without trajectory noise, and the finite-trajectory study finally replaces

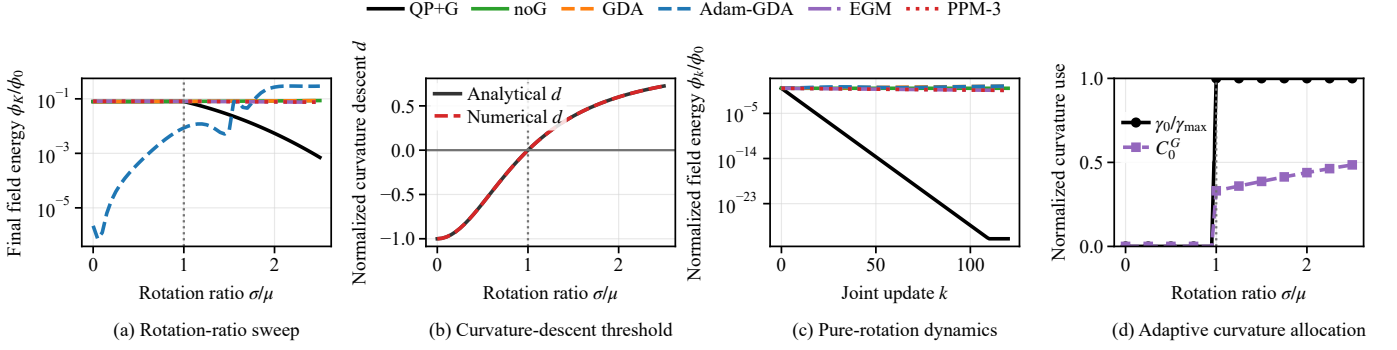


Fig. 2. Controlled normal-form results: (a) final normalized field energy; (b) analytical and computed curvature descent; (c) pure-rotation trajectories; and (d) first-update normalized curvature step size and relative curvature contribution. Vertical lines in (a), (b), and (d) mark $\sigma/\mu = 1$.

exact expected directions by sampled stochastic oracles. This progression isolates geometric adaptation, nonlinear policy parameterization, and sampling error in successive levels of evaluation. Methods are the proposed box-QP controller with curvature (QP+G), its field-only ablation noG ($\gamma_k = 0$), GDA, Adam-GDA [31], EGM, and the three-inner-iteration PPM approximation PPM-3. All methods update both players simultaneously; unless noted, fixed step sizes and QP caps are 0.03, $\eta_\psi = 1$ gives $\mathcal{E}_F = \phi$, and population normalizers are frozen at initialization with floor 10^{-10} . For every Markov-game experiment, $(\alpha, \varepsilon) = (0.9, 0.03)$; the regularized and unregularized value tolerances are 10^{-10} and 10^{-12} , and the spectral floor is 10^{-8} . Across matched seeds, all paired 95% confidence intervals are two-sided Student- t intervals for metric-oriented differences, with positive values favoring QP+G.

A. Controlled Normal-Form Geometry

We test the mechanism on the exactly solvable normal linear field with $\lambda_P = 0$, so Eq. (11) retains only field energy. For radial contraction rate $\mu \geq 0$ and angular rotation rate $\sigma \geq 0$,

$$\mathbf{F}(\mathbf{z}) = \mathbf{A}\mathbf{z}, \quad \mathbf{A} = \mu\mathbf{I}_2 + \sigma\mathbf{J}_0, \quad \mathcal{V}(\mathbf{z}) = \frac{1}{2}\|\mathbf{F}(\mathbf{z})\|^2.$$

Here $\mathbf{S} = \mu\mathbf{I}_2$, $\mathbf{W} = \sigma\mathbf{J}_0$, and direct calculation gives

$$e = \langle \nabla \mathcal{V}, \mathbf{F} \rangle = \mu(\mu^2 + \sigma^2) \|\mathbf{z}\|^2, \\ d = -\langle \nabla \mathcal{V}, \mathbf{G} \rangle = (\mu^2 + \sigma^2)(\sigma^2 - \mu^2) \|\mathbf{z}\|^2. \quad (35)$$

Eq. (35) shows that $+\mathbf{G}$ decreases the field energy exactly when $|\sigma| > |\mu|$, matching Theorem 1; moreover, $\max\{\chi_F, \chi_G\}/\Phi = \max\{\mu, \sigma^2 - \mu^2\}$ evaluates the coverage modulus in Eq. (28). For this quadratic field-energy function, the two-dimensional Taylor model is exact.

To quantify accepted curvature use, define its relative contribution as $C_k^G \triangleq \gamma_k \|\mathbf{G}(\mathbf{z}_k)\| / \max\{\beta_k \|\mathbf{F}(\mathbf{z}_k)\| + \gamma_k \|\mathbf{G}(\mathbf{z}_k)\|, 10^{-15}\}$. Fig. 2 sweeps $\sigma/\mu \in [0, 2.5]$; panels (a)–(c) recover the threshold and pure-rotation decay, while panel (d) shows first-update $\gamma_0/\gamma_{\max}$ and C_0^G from the common initial point. These quantities measure curvature step-size activation and the curvature share of the additive nominal update magnitude; their threshold-like profile follows here from exact directions and $\lambda_P = 0$.

B. Tabular Markov Games with Population Oracles

We span a one-state normal-form benchmark and sequential finite-state games while removing function approximation,

using rock–paper–scissors (RPS), five-state CyclicControl, and eight-state FrequencyHopping, each with three actions per player and tabular softmax logits. RPS is a standard multi-agent RL benchmark [32]; Appendix B specifies the games. For both the tabular and neural population experiments, the QP+G/noG controller uses the Markov-game composite Lyapunov function

$$V_{\text{MG}}(\mathbf{z}) = 0.3 \frac{\frac{1}{2}\|\mathbf{F}(\mathbf{z})\|^2}{\kappa_F} + \frac{\text{Gap}_\varepsilon(\mathbf{z})}{\kappa_P}, \quad (36)$$

This corresponds to $\lambda_F = 0.3$ and $\lambda_P = 1$, with both normalizers frozen at initialization. Population automatic differentiation computes $\mathbf{F}$ as the saddle gradient of J_ε and $\mathbf{G} = \nabla \mathbf{F} \mathbf{F}$, each stencil recomputes both regularized BRs, and Eq. (32) holds at every checkpoint.

Across ten initializations and 100 joint updates, QP+G raises worst-case return and lowers exploitability relative to noG in all three games. Exact one-sided paired sign tests of the return gains remain significant after Holm correction across the three games [33]. Fig. 3 shows that the same ordering extends to the regularized gap and field norm, linking the performance gain to improved saddle-field optimization.

C. Neural-Policy Markov Games with Population Oracles

The neural-policy experiment isolates nonlinear parameterization from sampling noise using separate 4–8–3 tanh–softmax policies, forming a 134-dimensional population field on the four games in Appendix B. An exact differentiable value-system solve supplies $\mathbf{F}$ and $\mathbf{G} = \nabla \mathbf{F} \mathbf{F}$; QP+G and noG evaluate Eq. (36) on nine- and three-point central stencils, respectively, with $\delta_D = \min\{10^{-2}, 10^{-3}/\max(\|\mathbf{D}\|, 10^{-12})\}$, $\mathbf{D} \in \{\mathbf{F}, \mathbf{G}\}$. Five screening seeds select each baseline step size from $\{0.001, 0.003, 0.01, 0.03\}$; ten disjoint test seeds use 60 simultaneous updates. QP+G attains higher worst-case return and lower exploitability than noG in all four games, with positive lower endpoints in all paired 95% confidence intervals for return gain; Fig. 4 displays three games, and SecurityPatrol has the same mean ordering. CyclicControl has the largest exploitability reduction, 79.1%. The game-wise mean of C_k^G ranges from 0.177 to 0.587. With $\mathbf{S}_k \triangleq \text{sym}(\nabla \mathbf{F}(\mathbf{z}_k))$ and $\mathbf{W}_k \triangleq \text{skew}(\nabla \mathbf{F}(\mathbf{z}_k))$, the median ratio $\|\mathbf{W}_k \mathbf{F}(\mathbf{z}_k)\| / (\|\mathbf{S}_k \mathbf{F}(\mathbf{z}_k)\| + 10^{-15})$ is 0.951 on FrequencyHopping and 0.966 on RoutingInterdiction. Theorem 1 characterizes the field-energy effect of $+\mathbf{G}$ alone,

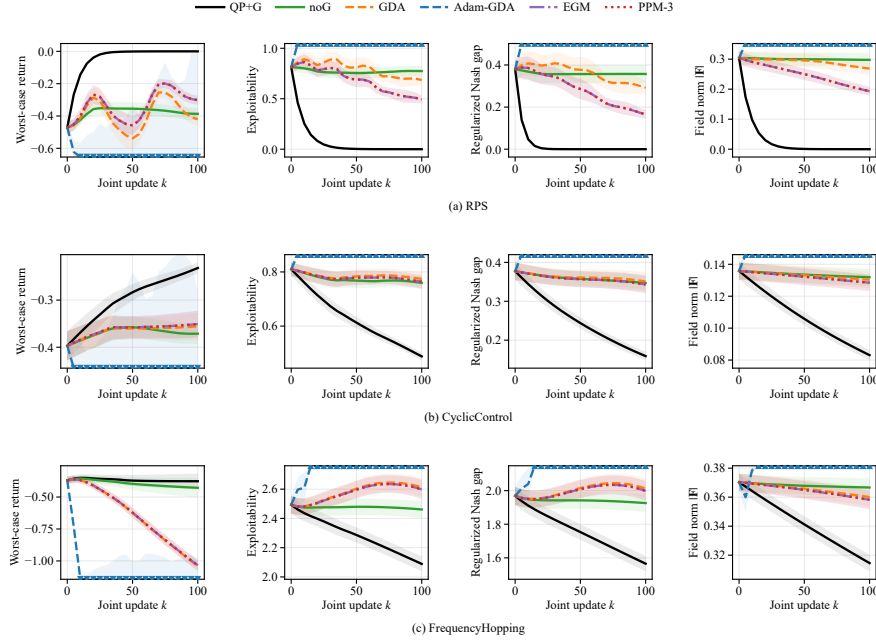


Fig. 3. Tabular population-oracle games over ten seeds (mean $\pm$ standard error); boundary triangles mark Adam-GDA means outside the displayed range.

whereas the composite QP can also activate curvature through the gap gradient and cross-curvature below the skew threshold.

D. Finite-Trajectory Oracle Validation

The preceding population-oracle studies evaluate the controller with exact expectations and therefore isolate its deterministic geometry. We now test whether the same two-direction mechanism remains observable when the field direction, curvature direction, and local-model stencil are all estimated from finitely many trajectories. At each of 60 CyclicControl updates, two 4–8–3 policies use 128 horizon-16 trajectories (2048 transitions). One detached-policy batch $\mathcal{B}_k$ supplies every stencil through unclipped per-decision prefix ratios and the infinitely differentiable Monte Carlo estimator (DiCE) [34]. DiCE gives the regularized field and Hessian–vector curvature oracles, with the field detached in the latter product. For $\lambda_P = 0$, an independent-batch-normalized field-energy Lyapunov function controls training, while population quantities are used only for checkpoint evaluation. Proposition 4 and Theorem 2 account for dependent same-batch stencil errors.

TABLE I
FINITE-TRAJECTORY CYCLICCONTROL AT $k = 60$ (2048 TRANSITIONS/UPDATE; TEN HELD-OUT SEEDS). ENTRIES ARE MEAN $\pm$ STANDARD ERROR WITH PAIRED 95% CONFIDENCE INTERVALS; POSITIVE Δ FAVORS QP+G.

Metric	QP+G	noG	Paired Δ
Worst-case return $\uparrow$	-0.203 ± 0.032	-0.546 ± 0.078	0.342 [0.123, 0.561]
Exploitability $\downarrow$	0.429 ± 0.047	0.932 ± 0.096	0.503 [0.274, 0.732]
Population $\ \mathbf{F}\ \downarrow$	0.169 ± 0.025	0.377 ± 0.037	0.207 [0.134, 0.281]

All paired intervals are positive; mean exploitability decreases by 54.0% relative to noG, and curvature remains active at every update. The diagnostics separate parameter-space stationarity ($\|\mathbf{F}\|$), unilateral policy improvement (exploitability),

and unregularized robust performance (worst-case return), while population checkpoints keep training noise out of evaluation. Thus the population-oracle curves test deterministic geometry, whereas Table I tests whether its benefits survive the biased, dependent sampled queries covered by the analysis; matched-seed intervals isolate the curvature-enabled controller.

VII. CONCLUSION

This paper developed complementary field and curvature directions for zero-sum Markov-game policy optimization. The symmetric–antisymmetric analysis identifies when curvature decreases field energy, while a closed-form box QP selects independent step sizes from a second-order Lyapunov-drift model. Under local geometric coverage and explicit stochastic-error budgets, the stationarity guarantee gives an $O(1/K)$ ergodic transient plus an error floor; with the local policy-space error bound, it transfers to Nash-gap and worst-case-return bounds. Population-oracle and finite-trajectory experiments support the predicted geometry and show improved worst-case return and exploitability over the field-only ablation. Independent step sizes thereby align curvature use with its local contribution to the composite Lyapunov model.

APPENDIX A PROOFS

a) *Proof of Proposition 1:* The GDA claim is immediate from Eq. (3). On the closed ball $\bar{B}(\mathbf{z}; r) \subset \mathcal{D}$, let M_i and B_F bound $\nabla^i \mathbf{F}$ ($i = 1, 2$) and $\mathbf{F}$. For $s \leq \min\{r/B_F, (2M_1)^{-1}, 1\}$, the contraction $T_s(\mathbf{u}) = \mathbf{z} - s\mathbf{F}(\mathbf{u})$ has PPM fixed-point displacement $\|\Delta\| \leq sB_F$. Taylor expansion yields $\Delta + s\mathbf{F} = -s\nabla\mathbf{F}\Delta - sr_2$, $\|r_2\| \leq M_2\|\Delta\|^2/2$, and hence $\Delta = -s\mathbf{F} + s^2\nabla\mathbf{F}\mathbf{F} + O(s^3)$ uniformly. For EGM, $\mathbf{F}(\mathbf{z} - s\mathbf{F}) = \mathbf{F} - s\nabla\mathbf{F}\mathbf{F} + O(s^2)$ uniformly; multiplication by $-s$ gives the same expansion. ■

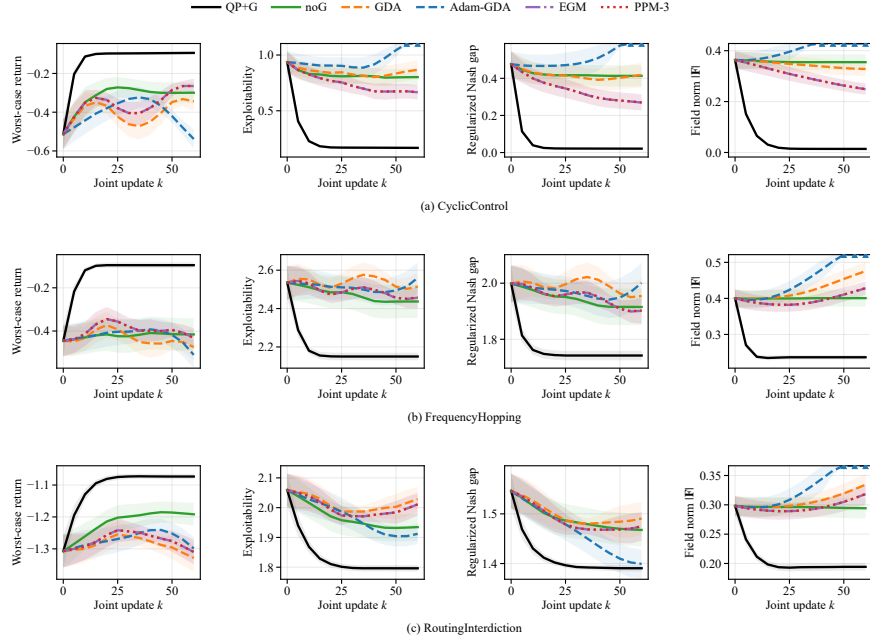


Fig. 4. Neural-policy population-oracle results over ten test seeds (mean $\pm$ standard error); triangles mark clipped Adam-GDA means.

b) Proof of Theorem 1: With $\mathbf{A} = \nabla \mathbf{F}$ and $\nabla \phi = \mathbf{A}^\top \mathbf{F}$, direct substitution gives Eq. (5) and $\langle \nabla \phi, \mathbf{G} \rangle = \mathbf{F}^\top \mathbf{A}^2 \mathbf{F}$. Expanding $\mathbf{A}^2$ as $\mathbf{S}^2 + \mathbf{S}\mathbf{W} + \mathbf{W}\mathbf{S} + \mathbf{W}^2$ eliminates the antisymmetric cross term, while $\mathbf{F}^\top \mathbf{S}^2 \mathbf{F} = \|\mathbf{S}\mathbf{F}\|^2$ and $\mathbf{F}^\top \mathbf{W}^2 \mathbf{F} = -\|\mathbf{W}\mathbf{F}\|^2$, proving Eq. (6). The right-hand side of Eq. (6) is negative exactly under Eq. (7). ■

c) Proof of Proposition 2: Positive definiteness makes $\hat{q}_k$ strictly convex; solving $\widehat{\mathbf{H}}_k(\beta, \gamma)^\top = (\hat{e}_k, \hat{d}_k)^\top$ gives Eq. (18) and its explicit inverse gives Eq. (19). An uncapped boundary solution lies on an axis, giving the two scalar candidates. For the capped problem, any boundary solution lies on one of the four edges; minimizing the resulting one-dimensional quadratic and clipping its stationary point gives the four displayed edge candidates. Strict convexity and direct objective comparison establish uniqueness and the comparator claim. ■

d) Proof of Proposition 3: The comparator sets are subsets of $[0, \infty)^2$, so set inclusion and scalar minimization prove dominance and their formulas. Completing the square in Eq. (19) gives $\hat{\Delta}_k^* = \hat{e}_k^2/(2\hat{c}_k) + (\hat{b}_k\hat{e}_k + \hat{c}_k\hat{d}_k)^2/(2\hat{c}_k\hat{d}_k)$. Subtracting $[\hat{e}_k]_+^2/(2\hat{c}_k)$ proves Eq. (20) and its equality condition. ■

e) Proof of Proposition 6: For $\tilde{\mathbf{A}}(\mathbf{z}) \triangleq \mathbf{W}_V^{1/2} \mathbf{A}(\mathbf{z}) \mathbf{W}_V^{-1/2}$, define $\tilde{\mathbf{S}}(\mathbf{z}) \triangleq \text{sym}(\tilde{\mathbf{A}}(\mathbf{z}))$, $\tilde{\mathbf{W}}(\mathbf{z}) \triangleq \text{skew}(\tilde{\mathbf{A}}(\mathbf{z}))$, and

$$h(\mathbf{z}, \mathbf{u}) \triangleq \max \left\{ \mathbf{u}^\top \tilde{\mathbf{S}}(\mathbf{z}) \mathbf{u}, \left\| \tilde{\mathbf{W}}(\mathbf{z}) \mathbf{u} \right\|^2 - \left\| \tilde{\mathbf{S}}(\mathbf{z}) \mathbf{u} \right\|^2 \right\}.$$

At $\mathbf{z}^*$, a unit $\mathbf{u}$ either has $\mathbf{u}^\top \tilde{\mathbf{S}}(\mathbf{z}^*) \mathbf{u} > 0$, or positive semidefiniteness gives $\tilde{\mathbf{S}}(\mathbf{z}^*) \mathbf{u} = 0$ and nonsingularity gives $\left\| \tilde{\mathbf{W}}(\mathbf{z}^*) \mathbf{u} \right\| = \left\| \tilde{\mathbf{A}}_* \mathbf{u} \right\| > 0$. Hence $h > 0$ on the compact unit sphere, so continuity yields $c_* > 0$ and a neighborhood $\mathcal{N}_*$ where $h \geq c_*$ uniformly. For $\Phi(\mathbf{z}) > 0$, set $\mathbf{u} = \mathbf{W}_V^{1/2} \mathbf{F}(\mathbf{z}) / \sqrt{\Phi(\mathbf{z})}$; then $\|\mathbf{u}\| = 1$, $\chi_F / \Phi = \mathbf{u}^\top \tilde{\mathbf{S}} \mathbf{u}$, and $\chi_G / \Phi = \left\| \tilde{\mathbf{W}} \mathbf{u} \right\|^2 - \left\| \tilde{\mathbf{S}} \mathbf{u} \right\|^2$, which proves the result. ■

f) Proof of Proposition 4: For Part (i), subtract the models and use $\hat{c}_k = \tilde{c}_k + \rho_{c,k}$ and $\hat{a}_k = \tilde{a}_k + \rho_{a,k}$. In $q_k - \hat{q}_k$ the shifts are nonpositive; dropping them and applying the five coefficient-error bounds gives the first inequality in Eq. (22), while reversing the difference gives the comparator bound. For Part (ii), Eq. (15), Taylor's bound, and the definitions of $\varepsilon_k^{\text{model}}$ and $\varepsilon_k^{\text{step,box}}$ give Eq. (23). For Part (iii), the step-size caps and $(a+b)^3 \leq 4(a^3 + b^3)$ imply $\varepsilon_k^{\text{rem}} \leq (2M_{3,k}/3)(\beta_{\max} + \gamma_{\max})^3 (\left\| \hat{\mathbf{F}}_k \right\|^3 + \left\| \hat{\mathbf{G}}_k \right\|^3)$. Conditional expectation gives Eq. (24), and the uniform bounds give its stated specialization. ■

g) Proof of Proposition 5: For Part (i), fix $k \geq k_0$ and compare the box-QP solution with $(s, 0)$ when $D_F(\mathbf{z}_k)$ is the largest positive signal, with $(0, s)$ when $D_G(\mathbf{z}_k)$ is the largest positive signal, and with $(0, 0)$ when both signals are nonpositive, resolving ties deterministically. These comparator choices are $\mathcal{F}_k$ -measurable. For either nonzero comparator, box optimality and Proposition 4 bound the realized increment by the comparator model plus $\varepsilon_k^{\text{box}}$ and, respectively, $s\delta_{e,k} + s^2(\delta_{c,k} + \rho_{c,k})/2$ or $s\delta_{d,k} + s^2(\delta_{a,k} + \rho_{a,k})/2$. Conditional expectation of the linear term contributes at most $-s[\max\{D_F(\mathbf{z}_k), D_G(\mathbf{z}_k)\}]_+ + sG_V \bar{b}_D$, while $\mathbb{E}[\left\| \hat{\mathbf{F}}_k \right\|^2 | \mathcal{F}_k] \leq M_F$ and $\mathbb{E}[\left\| \hat{\mathbf{G}}_k \right\|^2 | \mathcal{F}_k] \leq M_G$ bound the Hessian terms. When both signals are nonpositive, comparison with $(0, 0)$ gives $\mathcal{D}_k \leq \bar{R}_{\text{box}} \leq \mathcal{R}_D(s)$. Collecting the shift, coefficient-error, and residual budgets gives Eq. (26). For Part (ii), take total expectations in Eq. (26), sum from k_0 to $K-1$, and use $\mathcal{V} \geq 0$ to obtain Eq. (27). ■

h) Proof of Theorem 2: The positive part of the left side of Eq. (29) is no smaller than its right side. Average this inequality along the trajectory and apply Proposition 5. Rearranging the resulting inequality and using $\mathcal{R}_{\text{geo}}(s) = \mathcal{R}_D(s) + sr_P$ proves Eq. (30). ■

APPENDIX B

FINITE MARKOV-GAME ENVIRONMENTS

Let S denote the number of states; all games use actions $a, b \in \{0, 1, 2\}$ and a uniform initial state. For $S > 1$, $P(s' | s, a, b) = (1 - \xi) \mathbf{1}_{\{s' = T(s, a, b)\}} + \xi \mathbf{1}_{\{s' \neq T(s, a, b)\}} / (S - 1)$, where T , ξ , and $\mathbf{1}$ denote the nominal next-state map, transition-noise probability, and event indicator. Let $\varphi_s = 2\pi s / S$ and $M \triangleq \begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{bmatrix}$. RPS has $S = 1$, reward M_{ab} , and a deterministic self-loop; the remaining games are: *CyclicControl*: $S = 5$, $\xi = 0.02$, $T = (s + 1 + ((a - b) \bmod 3)) \bmod S$, and $r_s(a, b) = (1 + 0.12 \cos \varphi_s) M_{\pi_s(a), \pi_s(b)}$, where $\pi_s(j) = (j - s) \bmod 3$. *FrequencyHopping*: $S = 8$, $\xi = 0.08$, $T = (s + 1 + a - b) \bmod S$, $q_s(a) = 0.25 \cos(\varphi_s + 2\pi a / 3)$, and $r_s(a, b) = q_s(a) + 0.55$ if $a \neq b$, while $r_s(a, a) = q_s(a) - 1.10$. *RoutingInterdiction*: $S = 6$, $\xi = 0.10$, $T = (s + 1 + \mathbf{1}_{\{a=b\}} + a) \bmod S$, $c_s(a) = 1 + 0.20 \sin(\varphi_s + 2\pi a / 3)$, and $r_s(a, b) = c_s(a) - 0.55 - 1.35 \mathbf{1}_{\{a=b\}} - 0.15 \mathbf{1}_{\{a \neq b\}}$. *SecurityPatrol*: $S = 8$, $\xi = 0.08$, $T = (s + 1 + b + \mathbf{1}_{\{a \neq b\}}) \bmod S$, $v_s(b) = 0.8 + 0.25 \cos(\varphi_s + 2\pi b / 3)$, and $r_s(a, b) = v_s(b)$ if $a = b$, while $r_s(a, b) = -0.55 v_s(b)$ otherwise. Neural policies use $x_s = (\sin \varphi_s, \cos \varphi_s, u_s, 1)$, where u_s is the normalized state index $s / (S - 1)$ for *CyclicControl*, the action-wise range of $q_s(a)$ for *FrequencyHopping*, the action-wise standard deviation of $c_s(a)$ for *RoutingInterdiction*, and the action-wise mean of $v_s(b)$ for *SecurityPatrol*.

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